INCH-POUND

MIL-C-17/78C
13 April 1993
SUPERSEDING
MIL-C-17/78B
15 March 1977

MILITARY SPECIFICATION SHEET

CABLES, RADIO FREQUENCY, FLEXIBLE, COAXIAL, 50 OHMS, M17/78-RG217 AND M17/78-00001 (TEMPERATURE STABILIZED)

THIS CABLE USES PVC MATERIAL AND IS NOT TO BE USED IN AEROSPACE APPLICATIONS.

NOTE: THE AIR FORCE HAS RESTRICTED THE USE OF PVC IN AEROSPACE AND GROUND SUPPORT APPLICATIONS. CABLES WITH PVC JACKETING SHALL BE USED FOR RETROFIT PURPOSES ONLY UNTIL AN ALTERNATE JACKET IS APPROVED.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-17.

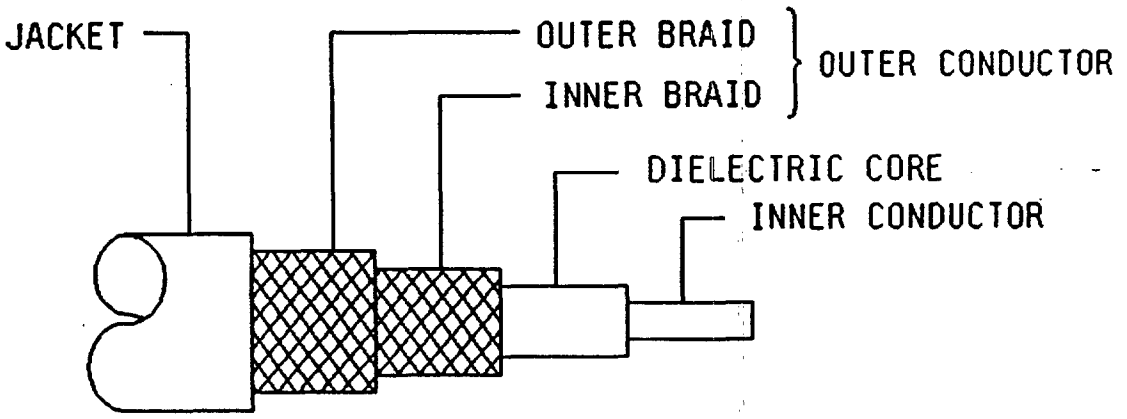


FIGURE 1. Configuration.

Changes to the text have been highlighted.

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TABLE I. Description.

Component	Construction Details
Inner conductor	Solid bare copper wire. Diameter: .106 inch ±.001.
Dielectric core	Type A-1: Solid polyethylene. Diameter: .370 inch ±.010.
Outer conductor	Double braid of AWG No. 33 bare copper wire. Diameter: .463 inch, maximum.
Inner braid	Coverage: 94.8 nominal Carriers: 24 Ends: 10 Picks/inch: 5.4 ±10%
Outer braid	Coverage: 93.6 nominal Carriers: 24 Ends: 8 Picks/inch: 10.6 ±10%
Jacket	Type IIa: PVC Diameter: .545 inch ±.010.

ENGINEERING INFORMATION:

Continuous working voltage: 5,800 V rms, maximum.

Operating frequency: 1 GHz maximum.

Velocity of propagation: 65.9 percent, nominal.

Power rating: See figure 2.

Operating temperature range: -40°C to +85°C.

Inner conductor properties:

DC resistance (maximum at 20°C): .0941 ohm per 100 feet.

Elongation: 30 percent, minimum.

Tensile strength: Not applicable.

Engineering note: This cable is useful in general purpose, medium low temperature applications. (See connector series "N" and "SC" per MIL-C-39012.)

REQUIREMENTS:

Dimensions, configuration, and description: See figure 1 and table I.

Environmental and mechanical:

Visual and mechanical examination:

Out-of-roundness: Applicable.

Eccentricity: 10 percent, maximum.

Adhesion of conductors:

Inner conductor to core: 15 pounds, minimum; 150 pounds, maximum.

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Aging stability: +98°C ±2°C.

Stress crack resistance: Not applicable.

Outer conductor integrity: Not applicable.

Cold bend: -40°C ±2°C.

Dimensional stability: +85°C ±2°C.

Inner conductor from core: .125 inch, maximum.

Inner conductor from jacket: .250 inch, maximum.

Temperature stabilization: (Applicable to M17/78-00001 cable only). 1/

Temperature cycle at -28°C within +0°C and -5°C, for 24 hours.

Temperature cycle at +65°C within +5°C and -0°C, for 24 hours.

Contamination: Applicable.

Bendability: Not applicable.

Flammability: Not applicable.

Weight: .225 pound per foot, maximum.

Electrical:

Continuity: Applicable.

Spark test: 8,000 V rms, +10%, -0%.

Voltage withstanding: 12,000 V rms, +10%, -0%.

Insulation resistance: Not applicable.

Corona extinction voltage: 7,000 V rms, minimum.

Characteristic impedance: 50 ±2 ohms.

Attenuation: See figure 2.

Structural return loss: See figure 3.

Capacitance: 32.2 pF per foot, maximum.

Capacitance stability: Not applicable.

Capacitance unbalance: Not applicable.

Transmission unbalance: Not applicable.

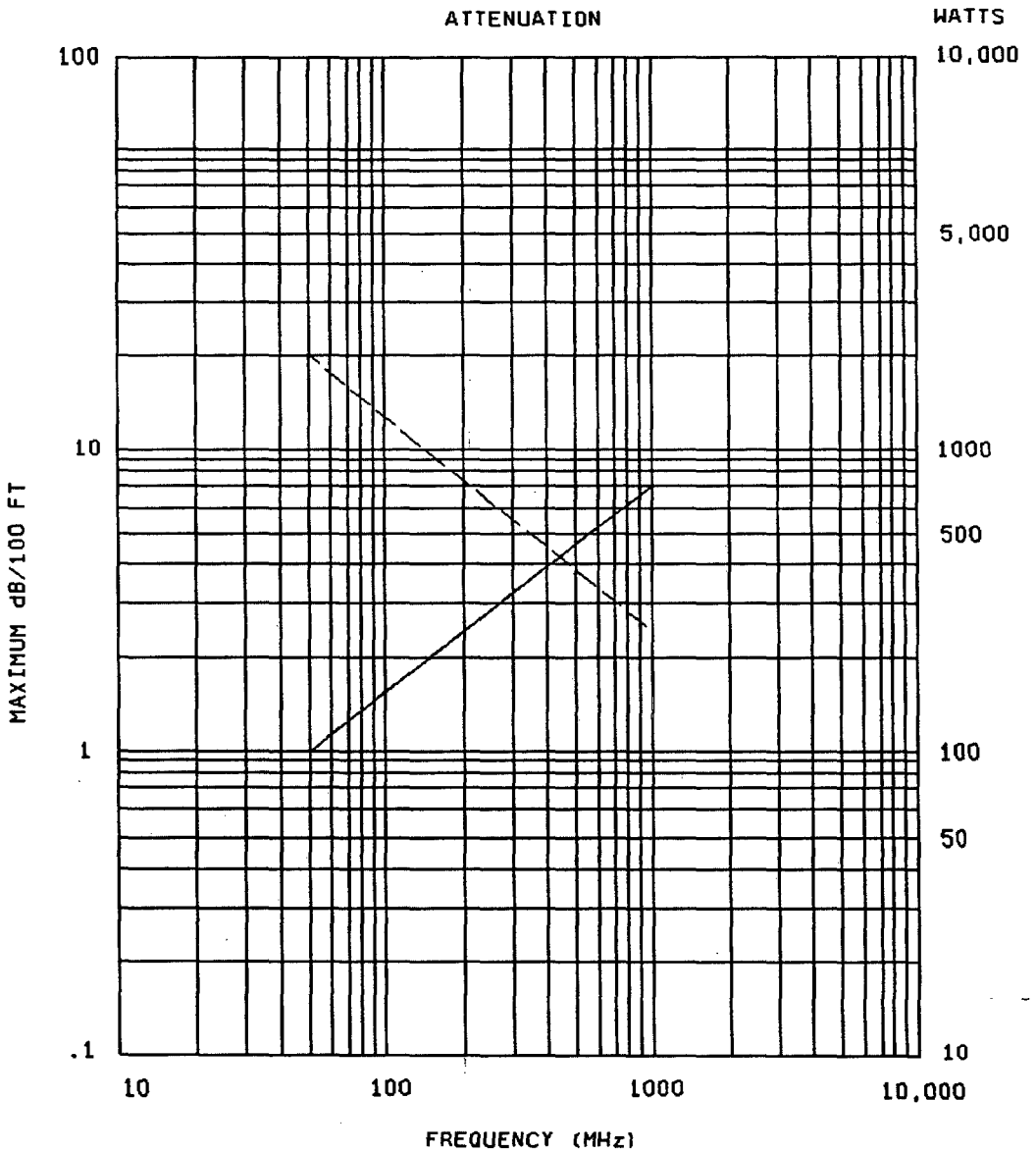
Mechanically induced noise voltage: Not applicable.

Time delay: Not applicable.

PIN (Part or identifying number): M17/78-RG217 and M17/78-00001.

1/ The cable (1000 feet) to be treated shall be wound on a spool having a 12 inch minimum barrel diameter before, during and after cycling. The temperature cycling shall be repeated 3 times. The time between cycling shall not exceed 30 minutes. This cable must be able to pass the dimensional stability test following temperature cycling.

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Power ----------

Attenuation _____

FIGURE 2. Attenuation.

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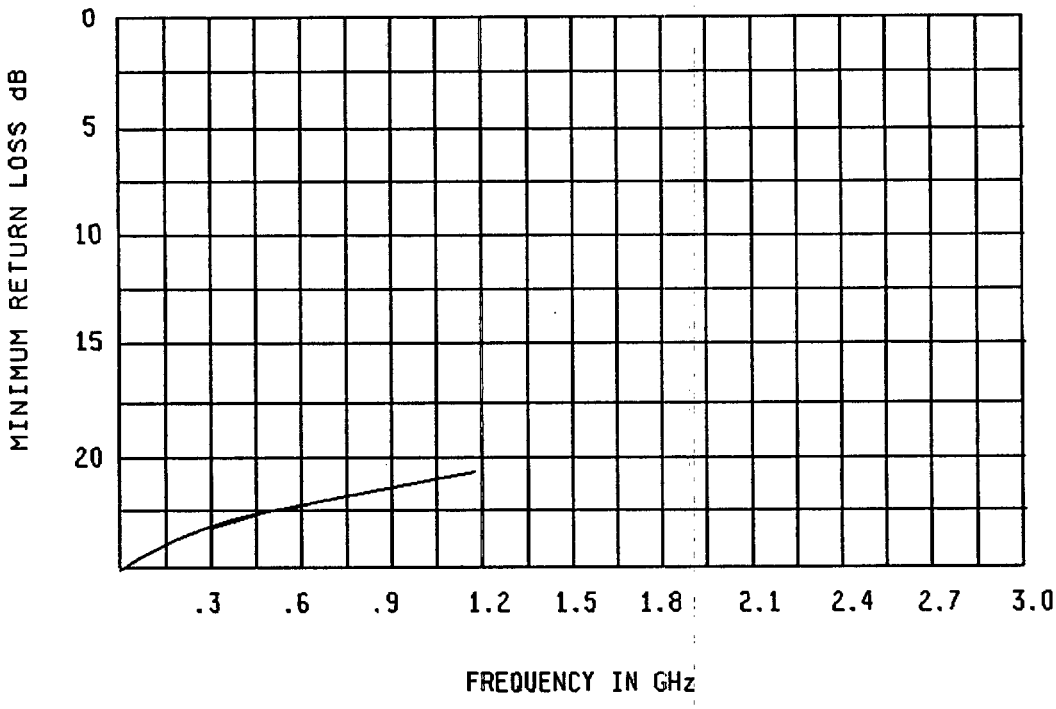


FIGURE 3. Structural return loss.

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Supersession: M17/78-00001 supersedes DESC drawing 92018 for temperature stabilized cable. When a QPL source is obtained the DESC drawing shall be inactivated.

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities:
Army - MI
Navy - SH, YD
Air Force - 17, 99
DLA - ES, IS

User activities:
Army - AR, AT, ME
Navy - AS, MC, OS
Air Force - 19

Preparing activity:
Army - CR

Agent:
DLA - ES

(Project 6145-2021-01)

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